SIMATS ENGINEERING

Department of Computer Science & Engineering

CO4-AT2 - Laboratory Performance | Programming in Java

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Course Code	CSA0904
Course Name	Programming in Java
Assignment	CO4-AT2 - Laboratory Performance (Question 3)

QUESTION 3: LAYOUT MANAGER SWITCHER (SWING)

> **Problem Statement:**
> Develop a Swing application that demonstrates the use of the following layout managers: FlowLayout, BorderLayout, and GridLayout. Switch among layouts using radio buttons.

SOURCE CODE

```
import java.awt.*;
import java.awt.event.ItemEvent;
import java.awt.event.ItemListener;
import javax.swing.*;

// Main Class at the Top
public class LayoutManagerSwitcher extends JFrame implements ItemListener {

    private final JPanel pnlCards;
    private final CardLayout cardLayout;
    private final JRadioButton rdoFlow;
    private final JRadioButton rdoBorder;
    private final JRadioButton rdoGrid;
    private final JLabel lblStatus;

    public LayoutManagerSwitcher() {
        setTitle("Interactive Layout Manager Switcher (Swing)");
        setSize(650, 480);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout(10, 10));

        // Top Control Panel: Radio Buttons
        JPanel pnlControls = new JPanel(new FlowLayout(FlowLayout.CENTER, 20, 10));
        pnlControls.setBackground(new Color(30, 60, 114));

        JLabel lblSelect = new JLabel("Select Layout:");
        lblSelect.setFont(new Font("Arial", Font.BOLD, 13));
        lblSelect.setForeground(Color.WHITE);
        pnlControls.add(lblSelect);

        rdoFlow = new JRadioButton("FlowLayout", true);
        rdoBorder = new JRadioButton("BorderLayout");
        rdoGrid = new JRadioButton("GridLayout (3x2)");
```

```
        // Grouping radio buttons for mutual exclusion
        ButtonGroup group = new ButtonGroup();
        group.add(rdoFlow);
        group.add(rdoBorder);
        group.add(rdoGrid);

        // Styling radio buttons
        for (JRadioButton rdo : new JRadioButton[]{rdoFlow, rdoBorder, rdoGrid}) {
            rdo.setForeground(Color.WHITE);
            rdo.setOpaque(false);
            rdo.setFocusPainted(false);
            rdo.addItemListener(this);
            pnlControls.add(rdo);
        }

        add(pnlControls, BorderLayout.NORTH);

        // Center Container with CardLayout to swap layout panels dynamically
        cardLayout = new CardLayout();
        pnlCards = new JPanel(cardLayout);

        // Add layout demonstration panels
        pnlCards.add(createFlowLayoutPanel(), "FLOW");
        pnlCards.add(createBorderLayoutPanel(), "BORDER");
        pnlCards.add(createGridLayoutPanel(), "GRID");

        add(pnlCards, BorderLayout.CENTER);

        // Footer Status Bar
        lblStatus = new JLabel(" Current Active Layout: FlowLayout (Components placed in a natural sequential flow)",
SwingConstants.CENTER);
        lblStatus.setBorder(BorderFactory.createCompoundBorder(
                BorderFactory.createMatteBorder(1, 0, 0, 0, Color.LIGHT_GRAY),
                BorderFactory.createEmptyBorder(8, 8, 8, 8)
        ));
        lblStatus.setFont(new Font("SansSerif", Font.PLAIN, 12));
        add(lblStatus, BorderLayout.SOUTH);
    }

    // 1. FlowLayout Demonstration Panel
    private JPanel createFlowLayoutPanel() {
        JPanel panel = new JPanel(new FlowLayout(FlowLayout.CENTER, 15, 25));
        panel.setBorder(BorderFactory.createTitledBorder("FlowLayout Preview"));
        panel.setBackground(new Color(245, 247, 250));

        String[] labels = {"Button 1", "Button 2", "Button 3", "Button 4", "Button 5"};
        for (String label : labels) {
            JButton btn = new JButton(label);
            btn.setPreferredSize(new Dimension(100, 40));
            btn.setBackground(new Color(41, 128, 185));
            btn.setForeground(Color.WHITE);
            panel.add(btn);
        }
        return panel;
    }

    // 2. BorderLayout Demonstration Panel
    private JPanel createBorderLayoutPanel() {
        JPanel panel = new JPanel(new BorderLayout(8, 8));
        panel.setBorder(BorderFactory.createTitledBorder("BorderLayout Preview"));
        panel.setBackground(new Color(245, 247, 250));

        JButton btnNorth = createStyledButton("NORTH Region", new Color(230, 126, 34));
        JButton btnSouth = createStyledButton("SOUTH Region", new Color(230, 126, 34));
        JButton btnEast = createStyledButton("EAST Region", new Color(39, 174, 96));
```

```
        JButton btnWest = createStyledButton("WEST Region", new Color(39, 174, 96));
        JButton btnCenter = createStyledButton("CENTER Region (Expands to Fill)", new Color(142, 68, 173));

        panel.add(btnNorth, BorderLayout.NORTH);
        panel.add(btnSouth, BorderLayout.SOUTH);
        panel.add(btnEast, BorderLayout.EAST);
        panel.add(btnWest, BorderLayout.WEST);
        panel.add(btnCenter, BorderLayout.CENTER);

        return panel;
    }

    // 3. GridLayout Demonstration Panel
    private JPanel createGridLayoutPanel() {
        JPanel panel = new JPanel(new GridLayout(3, 2, 10, 10));
        panel.setBorder(BorderFactory.createTitledBorder("GridLayout (3 Rows x 2 Columns) Preview"));
        panel.setBackground(new Color(245, 247, 250));

        Color[] colors = {
            new Color(52, 152, 219), new Color(46, 204, 113),
            new Color(155, 89, 182), new Color(241, 196, 15),
            new Color(231, 76, 60), new Color(52, 73, 94)
        };

        for (int i = 1; i <= 6; i++) {
            JButton btn = createStyledButton("Grid Cell " + i, colors[i - 1]);
            panel.add(btn);
        }
        return panel;
    }

    private JButton createStyledButton(String text, Color bg) {
        JButton btn = new JButton(text);
        btn.setBackground(bg);
        btn.setForeground(Color.WHITE);
        btn.setFont(new Font("Arial", Font.BOLD, 12));
        btn.setFocusPainted(false);
        return btn;
    }

    @Override
    public void itemStateChanged(ItemEvent e) {
        if (e.getStateChange() == ItemEvent.SELECTED) {
            if (e.getSource() == rdoFlow) {
                cardLayout.show(pnlCards, "FLOW");
                lblStatus.setText(" Current Active Layout: FlowLayout (Components placed in natural sequential flo
w)");
            } else if (e.getSource() == rdoBorder) {
                cardLayout.show(pnlCards, "BORDER");
                lblStatus.setText(" Current Active Layout: BorderLayout (Arranged into North, South, East, West, Cent
er)");
            } else if (e.getSource() == rdoGrid) {
                cardLayout.show(pnlCards, "GRID");
                lblStatus.setText(" Current Active Layout: GridLayout (Components placed into equal-sized rectangular
grid cells)");
            }
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> new LayoutManagerSwitcher().setVisible(true));
    }
}
```

OUTPUT SCREENSHOT